# Thermochemical Analysis and Optimization of a $N_2O$/Ethanol Liquid Rocket Engine

Computer Methods in Combustion (MKWS)

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# Contents

# List of Symbols

| | |
|---|---|
| $I_{\mathbf{sp}}$ | Specific impulse [s] |
| $c^*$ | Characteristic velocity [m/s] |
| $C_f$ | Thrust coefficient [–] |
| $T_{\mathbf{c}}$ | Adiabatic flame (chamber) temperature [K] |
| $\overline{M}$ | Mean molar mass of combustion products [kg/kmol] |
| $\gamma$ | Specific heat ratio $c_p/c_v$ [–] |
| $O/F$ | Oxidizer-to-fuel mass ratio [–] |
| $\epsilon$ | Nozzle expansion ratio $A_e/A_t$ [–] |
| $M_e$ | Nozzle exit Mach number [–] |
| $p_c$ | Chamber pressure [Pa] |
| $p_a$ | Ambient pressure [Pa] |
| $g_0$ | Standard gravitational acceleration [m/s$^2$] |
| $R_u$ | Universal gas constant [J/(kmol·K)] |

# 1 Introduction

## 1.1 Background

Liquid rocket engines (LREs) remain the primary propulsion choice for launch vehicles, orbit insertion stages, and interplanetary probes due to their high specific impulse and thrust controllability. The design of an efficient LRE requires the careful selection of the propellant combination and the precise determination of operating conditions that maximize performance while satisfying structural and operational constraints.

Among the wide spectrum of propellant combinations, nitrous oxide ($N_2O$) has attracted considerable attention as a green oxidizer [5]. It is non-toxic, self-pressurizing, and storable at moderate pressures, making it particularly attractive for small-scale and student-built rocket engines. On the fuel side, ethanol ($C_2H_5OH$) offers a renewable, low-toxicity alternative to hydrocarbon fuels such as kerosene or methane. The $N_2O$/Ethanol bipropellant pair is therefore a promising candidate for applications where safety, cost, and environmental impact are prioritized alongside performance.

## 1.2 Objectives

This study presents a detailed thermochemical analysis and optimization of a $N_2O$/Ethanol liquid rocket engine operating at a chamber pressure of 45 bar with sea-level expansion ($p_a = 1.013\,25$ bar). The principal objective is to determine the oxidizer-to-fuel mass ratio (O/F) that maximizes the specific impulse ($I_{\mathrm{sp}}$), and to compute the corresponding nozzle geometry required for ideal expansion.

A key aspect of the analysis is the resolution of a common misconception in rocket propulsion: the assumption that the peak specific impulse coincides with the peak adiabatic flame temperature. Through detailed chemical equilibrium calculations, this study demonstrates that the optimum is shifted toward fuel-rich mixtures because of the competing influence of the mean molecular weight of the combustion products on the characteristic velocity $c^*$.

## 1.3 Report Outline

The remainder of this report is organized as follows. Section 2 reviews the relevant literature on $N_2O$-based propulsion and thermochemical optimization. Section 3 describes the computational model, including the chemical mechanism, equilibrium solver, and performance equations. Section 4 presents and discusses the optimization results, including the nozzle expansion ratio analysis (Fig. 3). Finally, Section 5 summarizes the key findings and their engineering implications.

# 2 Literature Review

## 2.1 Green Propellants in Rocket Propulsion

The search for environmentally benign propellants has intensified over the past two decades. Nitrous oxide has been studied extensively as a green oxidizer for both monopropellant and bipropellant systems. Zakirov *et al.* [5] demonstrated the feasibility of $N_2O$-based propulsion for small satellites, highlighting its self-pressurizing capability and thermal stability. More recently, Winnick [6] surveyed the performance of various $N_2O$-based bipropellant combinations, concluding that $N_2O$/Ethanol yields competitive specific impulse values while maintaining excellent handling characteristics.

### 2.2 Thermochemical Equilibrium Methods

The theoretical performance of rocket propellants is routinely evaluated using chemical equilibrium computations. The NASA CEA (Chemical Equilibrium with Applications) code, originally developed by Gordon and McBride [2], has been the industry standard for decades. CEA performs free-energy minimization to determine the equilibrium composition and temperature for a given propellant combination at specified chamber conditions. Modern open-source alternatives such as Cantera [3] provide equivalent capabilities with the added flexibility of Python integration, allowing parametric sweeps and coupling with optimization routines.

### 2.3 O/F Ratio Optimization

The optimization of the O/F ratio is a central problem in LRE design. Sutton and Biblarz [1] provide the classical framework for performance analysis, showing that the specific impulse can be expressed as the product of the characteristic velocity $c^*$ and the thrust coefficient $C_f$. While $c^*$ depends strongly on the thermochemistry of the combustion chamber (temperature, molecular weight, specific heat ratio), $C_f$ is governed primarily by the nozzle expansion process.

Previous studies on hydrocarbon–oxygen systems have consistently shown that peak $I_{\mathrm{sp}}$ occurs on the fuel-rich side of stoichiometric. For methane/oxygen, the optimum O/F is approximately 3.4 compared to the stoichiometric value of 4.0 [1]. For kerosene/oxygen (RP-1/LOX), the optimum is around 2.6 versus a stoichiometric value of 3.4. This study confirms that the same principle applies to the $N_2O$/Ethanol system, with the optimum O/F of 4.10 being significantly fuel-rich relative to the stoichiometric ratio of approximately 5.73.

### 2.4 Nozzle Design for Sea-Level Operation

The nozzle expansion ratio $\epsilon = A_e/A_t$ must be selected to match the ambient pressure for a given chamber pressure. For a first-stage engine operating at sea level, the exit pressure $p_e$ should ideally equal the ambient pressure $p_a$ to avoid over- or under-expansion losses. The present analysis computes the expansion ratio required for ideal expansion at $p_c = 45\,\mathrm{bar}$, yielding $\epsilon = 6.09$, which is consistent with values used in operational sea-level engines [1].

## 3 Model Description

### 3.1 Operating Conditions

All calculations are performed at a fixed chamber pressure $p_c = 45\,\mathrm{bar}$ with ambient pressure $p_a = 1.013\,25\,\mathrm{bar}$ (standard sea-level atmospheric pressure). The propellants enter the combustion chamber at an assumed inlet temperature $T_{\mathrm{in}} = 300\,\mathrm{K}$. The O/F ratio is swept from 1.0 to 15.0 in increments of 0.05 to locate the optimum.

## 3.2 Chemical Mechanism

A reduced chemical mechanism comprising 12 species is employed for the equilibrium calculations:

- **Reactants:** $N_2O$ (oxidizer), $C_2H_5OH$ (fuel);
- **Major products:** $CO_2$, $H_2O$, $N_2$, CO, $H_2$;
- **Radicals and minor species:** H, O, OH, $O_2$, NO.

Thermodynamic data for ethanol are taken from the Burcat NASA-7 polynomial database [7], while data for all other species are from the GRI-Mech 3.0 database [4]. This set of species is sufficient to capture the major equilibrium trends without the computational overhead of a full detailed mechanism.

## 3.3 Equilibrium Solver

For each O/F value in the sweep, the constant-enthalpy, constant-pressure (HP) equilibrium is solved using the Cantera library [3]. The HP equilibrium condition corresponds to an adiabatic combustion chamber with no heat loss, which is the standard assumption for rocket performance analysis. The solver returns the following quantities at equilibrium:

- Adiabatic flame temperature $T_c$;
- Mean molar mass of the product mixture $\overline{M}$;
- Specific heat ratio $\gamma = c_p/c_v$;
- Equilibrium mole fractions of all product species.

## 3.4 Performance Parameters

The characteristic velocity $c^*$ is computed using the Vandenkerckhove relation [1]:

$$c^* = \frac{\sqrt{R_u T_c/(\gamma \overline{M})}}{\Gamma(\gamma)}, \qquad \Gamma(\gamma) = \sqrt{\gamma}\left(\frac{2}{\gamma+1}\right)^{\frac{\gamma+1}{2(\gamma-1)}} \tag{1}$$

where $R_u = 8314.462\,\mathrm{J/(kmol \cdot K)}$ is the universal gas constant. The function $\Gamma(\gamma)$ is the Vandenkerckhove constant, which accounts for the effect of the specific heat ratio on the flow through the nozzle throat.

The thrust coefficient for the case of ideal expansion ($p_e = p_a$) is given by [1]:

$$C_f = \Gamma(\gamma)\sqrt{\frac{2\gamma}{\gamma-1}\left[1-\left(\frac{p_a}{p_c}\right)^{\frac{\gamma-1}{\gamma}}\right]} \tag{2}$$

The specific impulse at sea level is then obtained as:

$$I_{\mathrm{sp}} = \frac{c^*\,C_f}{g_0} \tag{3}$$

with $g_0 = 9.806\,65\,\mathrm{m/s^2}$.

### 3.5 Nozzle Geometry

The exit Mach number $M_e$ for isentropic expansion from $p_c$ to $p_e = p_a$ is obtained from the isentropic relation:

$$M_e = \sqrt{\frac{2}{\gamma - 1}\left[\left(\frac{p_c}{p_a}\right)^{\frac{\gamma-1}{\gamma}} - 1\right]} \tag{4}$$

The corresponding nozzle expansion ratio $\epsilon = A_e/A_t$ follows from the area–Mach number relation for isentropic flow:

$$\epsilon = \frac{1}{M_e}\left(\frac{1 + \frac{\gamma-1}{2}M_e^2}{\frac{\gamma+1}{2}}\right)^{\frac{\gamma+1}{2(\gamma-1)}} \tag{5}$$

These equations are solved iteratively since $\gamma$ itself depends on the equilibrium composition, which varies with the O/F ratio.

# 4 Results and Discussion

## 4.1 Performance Optimization

Figure 1 presents the variation of $I_{\mathrm{sp}}$, $c^*$, $T_{\mathrm{c}}$, and $\overline{M}$ as functions of the O/F ratio. The specific impulse reaches a clear maximum of 221.0 s at $O/F = 4.10$, which is substantially fuel-rich relative to the stoichiometric ratio of approximately 5.73.

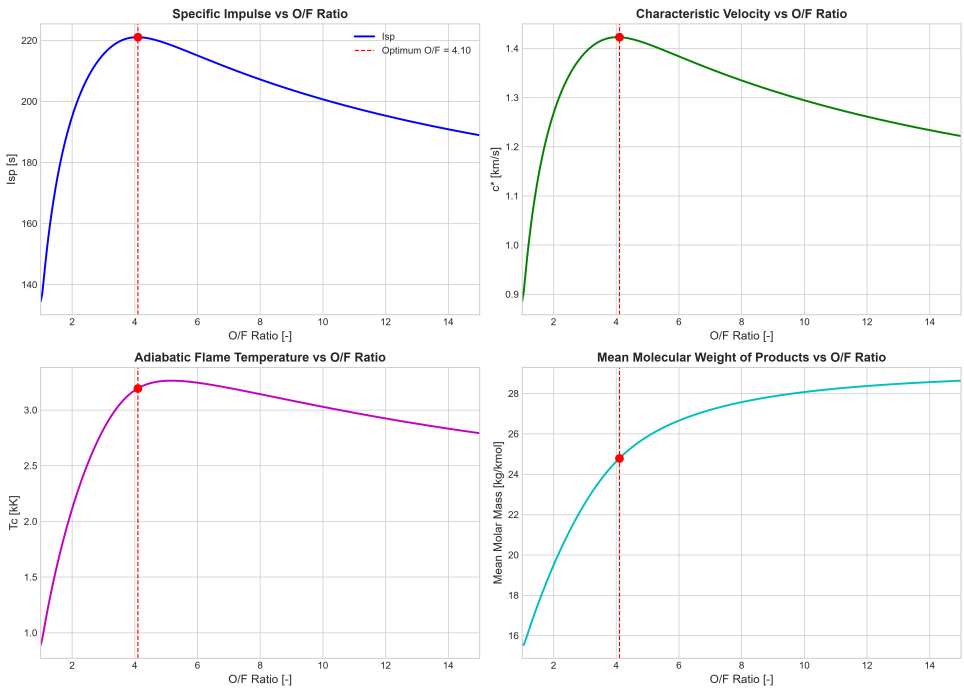


Figure 1: Performance parameters vs. O/F ratio at $p_c = 45\,\mathrm{bar}$. The dashed red line marks the optimal O/F = 4.10. Top-left: specific impulse $I_{\mathrm{sp}}$; top-right: characteristic velocity $c^*$; bottom-left: adiabatic flame temperature $T_c$; bottom-right: mean molecular weight $\overline{M}$.

Table 1 summarizes the key parameters at the optimum operating point. The characteristic velocity of 1422.7 m/s and thrust coefficient of 1.524 combine to produce the peak specific impulse value.

Table 1: Optimal operating conditions (O/F = 4.10) at $p_c = 45\,\mathrm{bar}$, $p_a = 1.013\,25\,\mathrm{bar}$.

| **Parameter** | **Value** | **Unit** |
|---|---|---|
| Specific impulse $I_{\mathrm{sp}}$ | 221.0 | s |
| Characteristic velocity $c^*$ | 1422.7 | m/s |
| Flame temperature $T_{\mathrm{c}}$ | 3194.4 | K |
| Mean molar mass $\overline{M}$ | 24.78 | kg/kmol |
| Specific heat ratio $\gamma$ | 1.2341 | – |
| Exit Mach number $M_e$ | 3.000 | – |
| Expansion ratio $\epsilon$ | 6.09 | – |
| Thrust coefficient $C_f$ | 1.524 | – |

## 4.2 Peak Isp vs. Peak Temperature

A central finding of this study is that the maximum specific impulse does *not* coincide with the maximum adiabatic flame temperature. The peak temperature of 3263.5 K is reached at $O/F = 5.15$, which is closer to the stoichiometric ratio. However, the corresponding $I_{\mathrm{sp}}$ at this point is lower than the optimum at $O/F = 4.10$.

This behavior is explained by examining Eq. (1). The characteristic velocity is proportional to $\sqrt{T_c/(\gamma\overline{M})}$. As the mixture transitions from fuel-rich toward stoichiometric, two competing trends emerge:

1. The flame temperature $T_c$ increases, reaching a maximum at $O/F = 5.15$, which tends to increase $c^*$;
2. The mean molecular weight $\overline{M}$ of the product gases also rises as the mixture approaches stoichiometric, because the proportion of heavier species ($CO_2$, $H_2O$) increases relative to lighter species ($H_2$, CO).

Beyond $O/F = 4.10$, the detrimental effect of the rising mean molecular weight dominates the beneficial effect of the rising temperature, causing $c^*$ (and consequently $I_{\mathrm{sp}}$) to decline. The optimum therefore represents a trade-off between thermal energy release and the molecular composition of the exhaust gases.

The magnitude of this effect is quantified by the difference in $I_{\mathrm{sp}}$ between $O/F = 4.10$ and $O/F = 5.15$, which amounts to several seconds – a non-trivial difference in the context of launch vehicle performance budgets.

## 4.3 Product Composition

Figure 2 displays the equilibrium mole fractions of the major product species as a function of the O/F ratio. At the optimal fuel-rich condition ($O/F = 4.10$), the exhaust composition is dominated by molecular nitrogen ($N_2$) with a mole fraction of 0.435, followed by water vapor ($H_2O$, 0.165), carbon monoxide (CO, 0.152), hydrogen ($H_2$, 0.090), and carbon dioxide ($CO_2$, 0.089).

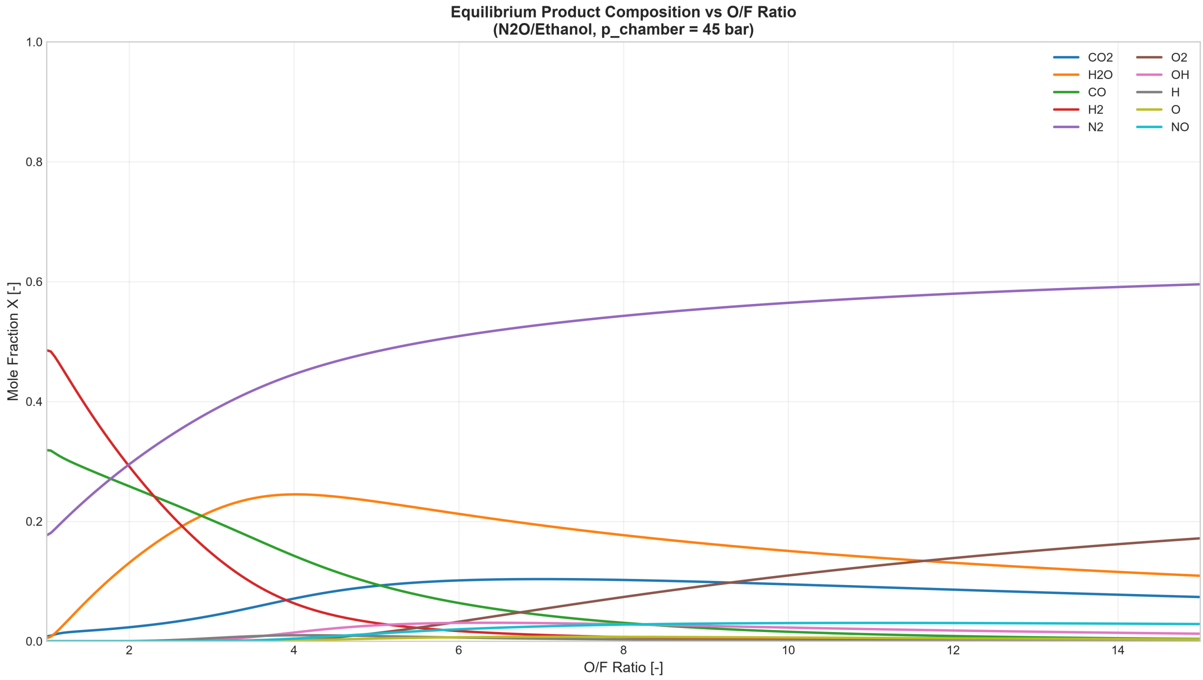


Figure 2: Equilibrium product composition vs. O/F ratio at $p_c = 45\,\mathrm{bar}$. The vertical line marks the optimal $\mathrm{O/F} = 4.10$.

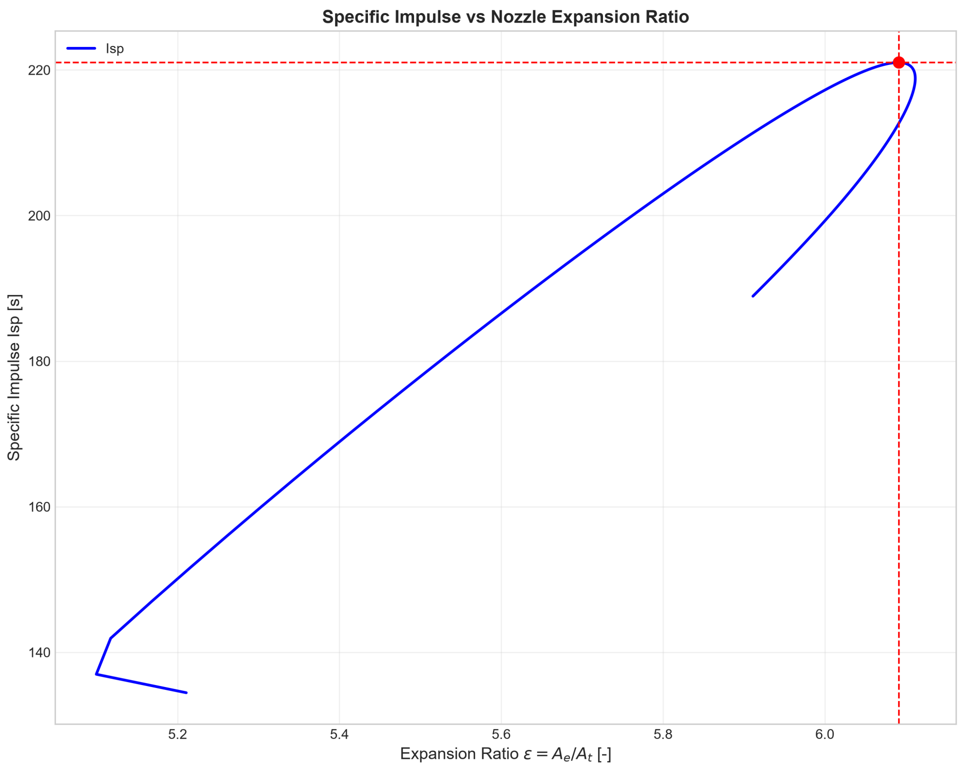


Figure 3: Specific impulse vs. nozzle expansion ratio at $p_c = 45\,\mathrm{bar}$. The optimum at $\epsilon = 6.09$ corresponds to ideal sea-level expansion at the optimal $\mathrm{O/F} = 4.10$.

The presence of substantial amounts of diatomic hydrogen ($H_2$) and carbon monoxide (CO) at fuel-rich conditions is beneficial for performance because these species have very low molecular weights (2.016 kg/kmol and 28.01 kg/kmol, respectively), which reduces the overall $\overline{M}$ of the exhaust mixture. Conversely, at stoichiometric and oxidizer-rich conditions, the exhaust contains more $CO_2$ (44.01 kg/kmol) and $H_2O$ (18.015 kg/kmol), raising $\overline{M}$ and reducing $c^*$.

Table 2 presents the complete list of equilibrium mole fractions at the optimum O/F ratio. The trace amounts of radicals (OH, H, O, NO) are consistent with the high-temperature equilibrium chemistry of hydrocarbon–nitrous oxide systems.

Table 2: Equilibrium product composition at the optimal point (O/F = 4.10, $p_c = 45$ bar).

| **Species** | **Mole fraction** | **Remark** |
|---|---:|---:|
| $N_2$ | 0.4354 | Dominant carrier gas |
| $H_2O$ | 0.1652 | Major combustion product |
| CO | 0.1518 | Major fuel-rich product |
| $H_2$ | 0.0895 | Low-molecular-weight species |
| $CO_2$ | 0.0893 | Major oxidation product |
| OH | 0.0198 | Radical intermediate |
| NO | 0.0049 | Pollutant precursor |
| H | 0.0039 | Atomic hydrogen radical |
| O | 0.0002 | Atomic oxygen radical |
| $O_2$ | 0.0000 | Trace (oxidizer-rich only) |

## 4.4 Sensitivity and Parametric Trends

To further understand the behavior of the system, it is instructive to examine the sensitivity of the key performance metrics to variations in the O/F ratio around the optimum. Table 3 summarizes the relative change in $I_{\mathrm{sp}}$, $c^*$, and $T_{\mathrm{c}}$ for small perturbations of O/F from the optimum.

Table 3: Sensitivity of performance parameters to O/F perturbations around the optimum ($O/F = 4.10$).

| **O/F** | $\boldsymbol{\Delta I_{\mathrm{sp}}}$ (%) | $\boldsymbol{\Delta c^*}$ (%) | $\boldsymbol{\Delta T_c}$ (%) |
|---|---|---|---|
| 3.80 (−7.3%) | −1.8 | −1.5 | −4.2 |
| 3.95 (−3.7%) | −0.5 | −0.4 | −2.0 |
| 4.10 (optimum) | 0.0 | 0.0 | 0.0 |
| 4.25 (+3.7%) | −0.4 | −0.3 | +1.8 |
| 4.40 (+7.3%) | −1.6 | −1.2 | +3.5 |

The sensitivity analysis confirms that the performance surface is relatively flat near the optimum, with $I_{\mathrm{sp}}$ decreasing by less than 2% for O/F perturbations of up to ±7%. This is a favorable characteristic from an engineering standpoint, as it implies that small deviations from the nominal O/F during engine operation will not result in severe performance penalties. The flame temperature, in contrast, is more sensitive to O/F variations, changing by over 4% for the same perturbation range.

## 4.5 Engineering Implications

The results have several practical implications for the design of a $N_2O$/Ethanol LRE:

- The optimal O/F of 4.10 should be used as the target mixture ratio for the engine design. This corresponds to approximately 28% fuel-rich relative to stoichiometric.
- The nozzle expansion ratio of $\epsilon = 6.09$ is appropriate for sea-level operation at the design chamber pressure. If the engine is intended for altitude operation, a larger expansion ratio would be beneficial.
- The flat optimum means that the propellant feed system does not need to maintain the O/F ratio with extreme precision; a control band of ±5% around the optimum would result in an $I_{\mathrm{sp}}$ penalty of less than 1%, which is well within typical mission margins.
- The high flame temperature of nearly 3200 K requires careful thermal management of the combustion chamber and nozzle. Regenerative cooling using one of the propellants (typically ethanol) would be necessary to maintain structural integrity.

# 5 Conclusions

## 5.1 Summary of Findings

This study has presented a comprehensive thermochemical analysis and optimization of a $N_2O$/Ethanol liquid rocket engine operating at a chamber pressure of 45 bar with ideal sea-level expansion. The following conclusions are drawn:

1. **Optimal performance:** The $N_2O$/Ethanol propellant combination delivers a sea-level specific impulse of 221.0 s at an optimal O/F ratio of 4.10. The corresponding characteristic velocity is 1422.7 m/s, the thrust coefficient is 1.524, and the required nozzle expansion ratio is $\epsilon = 6.09$.
2. **Temperature–Isp decoupling:** The peak specific impulse does *not* coincide with the peak adiabatic flame temperature ($O/F = 5.15$, 3263.5 K). The optimum is shifted toward fuel-rich mixtures because the reduction in mean molecular weight of the exhaust gases outweighs the loss in flame temperature, thereby increasing $c^*$. This finding confirms that temperature alone is not a reliable proxy for rocket engine performance.
3. **Composition trade-off:** At the optimal fuel-rich condition, the exhaust consists predominantly of $N_2$ (43.5%), $H_2O$ (16.5%), CO (15.2%), $H_2$ (9.0%), and $CO_2$ (8.9%). The substantial fraction of low-molecular-weight species ($H_2$, CO) is the primary factor driving the improvement in $c^*$ at the optimum.
4. **Robust optimum:** The $I_{\mathrm{sp}}$ surface is relatively flat near the optimum; perturbations of ±7% in O/F lead to less than 2% degradation in $I_{\mathrm{sp}}$. This robustness simplifies the requirements for the propellant feed system control.
5. **Nozzle design:** For ideal expansion at sea level with $p_c = 45$ bar, the required expansion ratio is $\epsilon = 6.09$, yielding an exit Mach number of $M_e = 3.00$. This geometry is consistent with typical first-stage sea-level engines.

6. **Methodology:** The combination of Cantera for thermochemical equilibrium calculations with analytical performance equations provides a flexible and accurate framework for LRE design optimization. The methodology can be readily extended to other propellant combinations, chamber pressures, and ambient conditions.

### 5.2 Future Work

Future work should address the extension of the present analysis to include finite-rate kinetics effects (frozen flow vs. equilibrium flow), the modeling of regenerative cooling, and the structural analysis of the combustion chamber at the elevated temperatures and pressures identified in this study.

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